VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI-590018

2020-2021

A

Mini Project Report

On

"FAKE NEWS DECTECTION USING MACHINE LEARNING"

Submitted in the partial fulfilment of the requirements for the VI Semester project-
18ECMP68

Bachelor of Engineering

In

Electronics and Communication Engineering

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CERTIFICATE

Certified that the Mini Project Work entitled ***"FAKE NEWS DECTECTION USING MACHINE LEARNING"*** is a bonafide work carried out by *AJAZ AHMED I-1GV18EC002, AWAIZ PASHA-1GV18EC003, KRITISH KUMAR R - 1GV18EC014, RAKESH A S -1GV18EC029* in the partial fulfillment for the award of degree of Bachelor of Engineering in *Electronics and Communication Engineering of the Visvesvaraya Technological University*, Belagavi during the year 2020-2021. It is certified that all corrections/suggestions indicated for the assessment have been incorporated in the report deposited in the departmental library. Mini Project report has been approved as it satisfies the academic requirement in respect of Mini Project work **-18ECMP68** prescribed for the Bachelor of Engineering Degree.

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ACKNOWLEDGEMENT

It is with deep feeling of gratitude we would like to express our sincere thanks to our institution **Dr. T. THIMMAIAH INSTITUTE OF TECHNOLOGY, K.G.F** for providing excellent infrastructure for the completion of the

We wish to express a wholehearted thanks to our **Principal Dr. Syed Ariff** for providing good infrastructure for undertaking this in college.

We would like to extend hearty thanks to our **Dean (Academic) Prof. Ruckmani Divakaran,** for being constant support of encouragement to carry out the mini project work.

We would like to extend hearty thanks to our **HOD and guide Prof. Vijaya Bharathi M,** for her timely support and guidance in the completion of this mini project work.

We would like to thank project coordinator **Dr. Jenitha A, Associate professor**, for her timely support and co-ordination in the completion of this mini project work.

We would like to thank all teaching and non-teaching staff who were directly and indirectly supported for carrying out this mini project work.

We extend our hearty thanks to our parents, friends for all the moral support provided during the preparation for the mini project work.

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SYNOPSIS

Newspapers are the primary source of news for people worldwide. However, due to the significant growth and updates in technologies, there has been a stupendous rise in the popularity of social media. The number of people who use social media has increased remarkably. As a consequence, social networks such as social media, websites, blogs, etc. have emerged as relevant platforms to gather all kinds of news. People rely more on social networks than newspapers these days. With the availability of the internet, these networks can be accessed easily. This can lead to easy manipulation of the existing news, thereby causing fake news. Fake news can be used as a vital tool to project people in a wrong way. It can spread hate among people which can further harm the society. Hence, it is very necessary to prevent the spread of fake news.

The news websites are publishing the news and provide the source of authentication. The question is how to authenticate the news and articles which are circulated among social media. It is harmful for the society to believe on the rumors and pretend to be a news. The need of an hour is to stop the rumors especially in the developing countries like India, and focus on the correct, authenticated news articles. With the help of Machine learning, author tried to aggregate the news and later determine whether the news is real or fake using machine learning algorithms. With The Help of Machine Learning We Tried to Aggregate the News and Later Determined Whether the News Is Real or Fake Using Logistic Regression, Decision Tree Classifier, Gradient Boosting Classifier, Random Forest Classifier.

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Chapter 1

INTRODUCTION

As an increasing amount of our lives is spent interacting online through social media platforms, more and more people tend to seek out and consume news from social media rather than traditional news organizations. The reasons for this change in consumption behaviours are inherent in the nature of these social media platforms:

(i) it is often more timely and less expensive to consume news on social media compared with traditional news media, such as newspapers or television; and

(ii) it is easier to further share, comment on, and discuss the news with friends or other readers on social media. For example, 62 percent of U.S. adults get news on social media in 2016, while in 2012, only 49 percent reported seeing news on social media1. It was also found that social media now outperforms television as the major news source2. Despite the advantages provided by social media, the quality of news on social media is lower than traditional news organizations. However, because it is cheap to provide news online and much faster and easier to disseminate through social media, large volumes of fake news, i.e., those news articles with intentionally false information, are produced online for a variety of purposes, such as financial and political gain. It was estimated that over 1 million tweets are related to fake news "Pizzagate"3 by the end of the presidential election. Given the prevalence of this new phenomenon, "Fake news" was even named the word of the year by the Macquarie dictionary in 2016. The extensive spread of fake news can have a serious negative impact on individuals and society. First, fake news can break the authenticity balance of the news ecosystem.

For example, it is evident that the most popular fake news was even more widely spread on Facebook than the most popular authentic mainstream news during the U.S. 2016 president election4. Second, fake news intentionally persuades consumers to accept biased or false beliefs. Fake news is usually manipulated by propagandists to convey political messages or influence. For example, some report shows that Russia has created fake accounts and social bots to spread false stories5. Third, fake news changes the way people interpret and respond to real news. For example, some fake news was just created to trigger people's distrust and make them confused, impeding their abilities to differentiate what is true from what is not6. To help mitigate the negative effects caused by fake news–both to benefit the public and the news ecosystem–It's critical that we develop methods to automatically detect fake news on social media.

1.1 History of the Project

In the modern era, the spread of fake news has become very evident. Fake news is being used for both economic and political benefits. The need of the hour is to prevent the spread of fake news. The first thing that needs to be done to achieve this is to detect fake news. Our project aims to develop a machine learning program to identify when a news source may be producing fake news. We use a corpus of labelled real and fake articles to build a classifier that can make decisions about information based on the content from the corpus. Our model focuses on identifying sources of fake news, based on multiple articles originating from a source. Once a source is labelled as a producer of fake news, we predict that all future articles from the same source are also a producer of fake news. The intended application of our project is to assist in applying visibility weights in social media. Social networks can make use of the weights produced by the model to obscure stories that are highly likely to be fake news.

As per authors in [13] the Information fabrication is not new. The authors have referred to columnist of Guardian Natalie Nougayrède who says “ The use of propaganda is ancient, but never before has there been the technology to so effectively disseminate it”. Falsified information, distorted information propaganda based information and fun based false information have been human communication features since Roman times. The impact and penetration of social media have dramatically changed reach of falsified information. The introduction of smart gadgets and very low-cost internet cost have added to its reach. In India even the remotest village has access to smart phones and internet services. Although there are numerous advantages of these services but it comes at a cost, the cost of rapid dissemination of falsified information along with substantiate information. Last 10 years have witnessed manifold increase in the number of users on the social media and microblogging. The data/text available on these sites in the form of news, blogs, posts , reviews, opinions, suggestions, arguments, comments etc. provides growth in the field of techniques and methods in the authenticity of these posts. Many researches have been conducted where machine learning have been used to automatically detect the fake news items. Also, few research works have been conducted using deep learning for auto feature extraction in fake news detectors.

1.2 Reason For Selecting The Project

Fake news been around for decades and is not a new concept. Nowadays social media is widely used everywhere where the rumours are spreading widely which has to be stopped.

Classification of any news item /post / blog into Fake or Real one has generated great interest from researchers. Many Scientists believe that Fake news issue may be addressed by means of machine learning and Artificial Intelligence.

This project shows a simple approach for fake news detection using machine learning algorithms. A different Machine Learning approaches have been attempted to detect it. Fake news is a global issue as well as a global challenge. In this project we can minimize the rumors spread through social media by detecting fake news. This has been a great reason for us to work on this project.

1.3 Requirements

1. **Computer:**(Operating System: Windows 10 With 64-bit, RAM:8gb and 2GHZ Processor)

2. **Data Sets:** True Data Set and Fake Data Set

3. **Coding Software:** Jupyter Notebook and Python 3.9

4. **Libraries:** Re, String, Pandas, Numpy, Sklearn.

Chapter 2

LITERATURE SURVEY

A literature survey or a literature review in a project is a type of review articles. It is a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literatures reviews are secondary sources, and do not report new or original experimental work. It is a basis for research in nearly every academic field. Concentrate on the own field of expertise.

2.1 Details of Literature:

Rohit Kumar Kaliyar (2019) proposed a Multi-class Counterfeit News Detection utilizing Ensemble Machine Learning of which they report about the use of Naive Bayes Algorithm and also, k-NN calculation which is a nonparametric strategy utilized for characterization and relapse and Decision Trees as well. They examined all aspects and tried to implement Gradient Boosting Machine which is a High-performance ML Algorithm which has significant achievement in the field of Machine Learning, which could yield around 86% accuracy and Confidence score of about 76%, And as number of iterations increase the training losses decrease iteratively. And as the number of iterations increases the time required will relatively decrease. These characteristics make the Gradient Boosting algorithm opt for the system.

Smitha.N (2020) proposed ML classifiers for Fake news recognition and its accuracy. In this proposed system they have referred to Count-Vectorizer, TF-IDF Vectorizer, Word Embedding for the calculation of best accuracy and best performance. By utilizing classification algorithms the accuracy obtained with SVM linear classification algorithm with TF-IDF Vectorizer feature extraction is of 94% accuracy.

They concluded that use of Neural networks has similar accurate results than SVM linear classification algorithms. As using Neural networks with linear classification algorithms, would make the classification more and more complex, Hence they opted the method of SVM over Neural Networks which is less complex. The first stage in the proposed system is text collection referred to datasets. Second step is text pre-processing in which conversion steps are included. Third is Feature extraction which includes encoding of words. Fourth is Classifiers of News which includes SVM, Logistic Regression, Decision Trees etc . Classification with TF-IDF Vectorizer yield more accurate results than Count-Vectorizer and Word Embedding process, Therefore they opted for TF-IDF Vectorizer for Classification.

Ranojoy Barua (2019) proposed An Application for Fake News Article Detection using Machine Learning Techniques. This system is developed based on machine learning approaches such as Long short term (LSTM) and Gated recurrent unit (GRU) to characterize a information into spam or original. The trial results on the dataset arranged in this work looks accurate. The model is additionally prepared and tried on other data and similar outcomes show the productivity of the proposed model. To detect the working of FNAD model. It is tried and utilized on 1,000 text information in FNAD's data to comprehend the execution of the model they have utilized estimation measurements for example, Confusion model, Score model and Accuracy model. Execution results of the proposed model using FNAD produces 80.2% accuracy. Hence it can be utilized to confirm text information from different sources of the web prior to tolerating it as a reality.

Chaitra K Hiramath (2019) proposed Fake News Detection Using Deep Learning Techniques in this model they have utilized classification algorithms for dividing fake and real information depending upon news sources. The algorithms utilized by this model are Support vector machine (SVM), Random forest (RF) and Deep Neural Network (DNN). The memory examination of different calculations shows that Logistic Regression needs less amount of memory space and the time examination of algorithms shows that Deep neural networks require 400 ms which is less time. The Accuracy calculation shows that Deep neural network has 91% accuracy which is greater than other four algorithms. So by using regression, Support vector machines (SVM), Random forest, Deep neural network (DNN) classification techniques it is easy, faster and accurate to classify fake news from a huge amount of datasets.

Vanya Tiwary (2020) proposed classification of News headlines using Machine Learning, Clickbait, Propaganda, Comment or Opinion and Satire or Humor are the classified types of Fake News. In this they have referred to Count Vectorizer, TF-IDF Vectorizer, Hashing Vectorizer. They proposed the implementation of Logical Regression, Decision Tree, Random Forest with different Vectorizers which gave the result that use of Logistic Regression algorithm with TF-IDF Vectorizer has more accurate results when compared to other algorithms with TF-IDF Vectorizer. Over 73% of headline Fake News was accurately detected and classified, whereas Decision Tree was accurate upto 62% and Random Forest upto 66% respectively. The research done by them was successful in classifying the HeadLines of News. By this proposal they figured out that a Logical Regression algorithm with TFIDF Vectorizer would yield the best results for the classification of Headline in the News.

Mykhailo Granik et. al. in their paper [3] shows a simple approach for fake news detection using naive Bayes classifier. This approach was implemented as a software system and tested against a data set of Facebook news posts. They were collected from three large Facebook pages each from the right and from the left, as well as three large mainstream political news pages (Politico, CNN, ABC News). They achieved classification accuracy of approximately 74%. Classification accuracy for fake news is slightly worse. This may be caused by the skewness of the dataset: only 4.9% of it is fake news.

Himank Gupta et. al. [10] gave a framework based on different machine learning approach that deals with various problems including accuracy shortage, time lag (BotMaker) and high processing time to handle thousands of tweets in 1 sec. Firstly, they have collected 400,000 tweets from HSpam14 dataset. Then they further characterize the 150,000 spam tweets and 250,000 non- spam tweets. They also derived some lightweight features along with the Top-30 words that are providing highest information gain from Bag-of-Words model. 4. They were able to achieve an accuracy of 91.65% and surpassed the existing solution by approximately18%.

Table 2.1: Literature survey of fake news detection

YEAR	AUTHOR	TITLE	DESCRIPTION	REMARK
2019	Chaitra K Hiramath	Fake News Detection Using Deep Learning Techniques.	In this model they have utilized classification algorithms for dividing fake and real information depending upon news sources. The algorithms utilized by this model are Support vector machine, Random forest and Deep Neural Network.	Accuracy Achieved Was 91%.
2019	Ranojoy Barua	Application for Fake News Article Detection using Machine Learning Techniques.	This system is developed based on machine learning approaches such as Long short term (LSTM) and Gated recurrent unit (GRU) to characterize a information into spam or original.	The Accuracy Achieved Was 80.2%.
2020	Smitha.N	ML classifiers for Fake news recognition and its accuracy	In this proposed system they have referred to Count-Vectorizer, TF-IDF Vectorizer, Word Embedding for the calculation of best accuracy and best performance.	The Accuracy Achieved Was 94%.

2019	Rohit Kumar Kaliyar	Multi-class Counterfeit News Detection utilizing Ensemble Machine Learning	They examined all aspects and tried to implement Gradient Boosting Machine which is a High-performance ML Algorithm which has significant achievement in the field of Machine Learning, which could yield around 86% accuracy.	The Accuracy Achieved Was 86%.
2020	Vanya Tiwary	Classification of News headlines using Machine Learning.	In this they have referred to Count Vectorizer. They proposed the implementation of Logical Regression, Decision Tree, Random Forest with different Vectorizers.	The Accuracy Achieved Was 73%.
2017	Mykhailo Granik	Fake news detection using naive Bayes classifier.	This approach was implemented as a software system and tested against a data set of Facebook news posts.	The Accuracy Achieved Was 74%.

2.2 Objective

To generate a model that can discriminate between “fake” and “true” news articles when it is trained with a certain dataset.

Chapter 3

METHODOLOGY

3.1 Introduction To Methodology

Methodology is the systematic, theoretical analysis of the methods applied to a field of study. It comprises the theoretical analysis of the body of methods and principles associated with a branch of knowledge. Typically, it encompasses concepts such as paradigm, theoretical model, phases and quantitative or qualitative techniques.

A methodology does not set out to provide solutions it is therefore, not the same as a method. Instead, a methodology offers the theoretical underpinning for understanding which method, set of methods, or best practices can be applied to a specific case, for example, to calculate a specific result. It has been defined as follows:

[1] The analysis of the principles of methods, rules, and postulates employed by a discipline.

[2] The systematic study of methods that are, can be, or have been applied within a discipline.

[3] The study or description of methods.

The Methodology is the general research strategy that outlines the way in which research is to be undertaken and, among other things, identifies the methods to be used in it. These methods, described in the methodology, define the means or modes of data collection or, sometimes, how a specific result is to be calculated. Methodology does not define specific methods, even though much attention is given to the nature and kinds of processes to be followed in a particular procedure or to attain an objective. When proper to a study of methodology, such processes constitute a constructive generic framework, and may therefore be broken down into sub-processes, combined, or their sequence changed.

Methodology and method are not interchangeable. In recent years, however, there has been a tendency to use methodology as a "pretentious substitute for the word method". Using methodology as a synonym for method or set of methods leads to confusion and misinterpretation and undermines the proper analysis that should go into designing research.

3.2 Overall Block Diagram of Fake News Detection

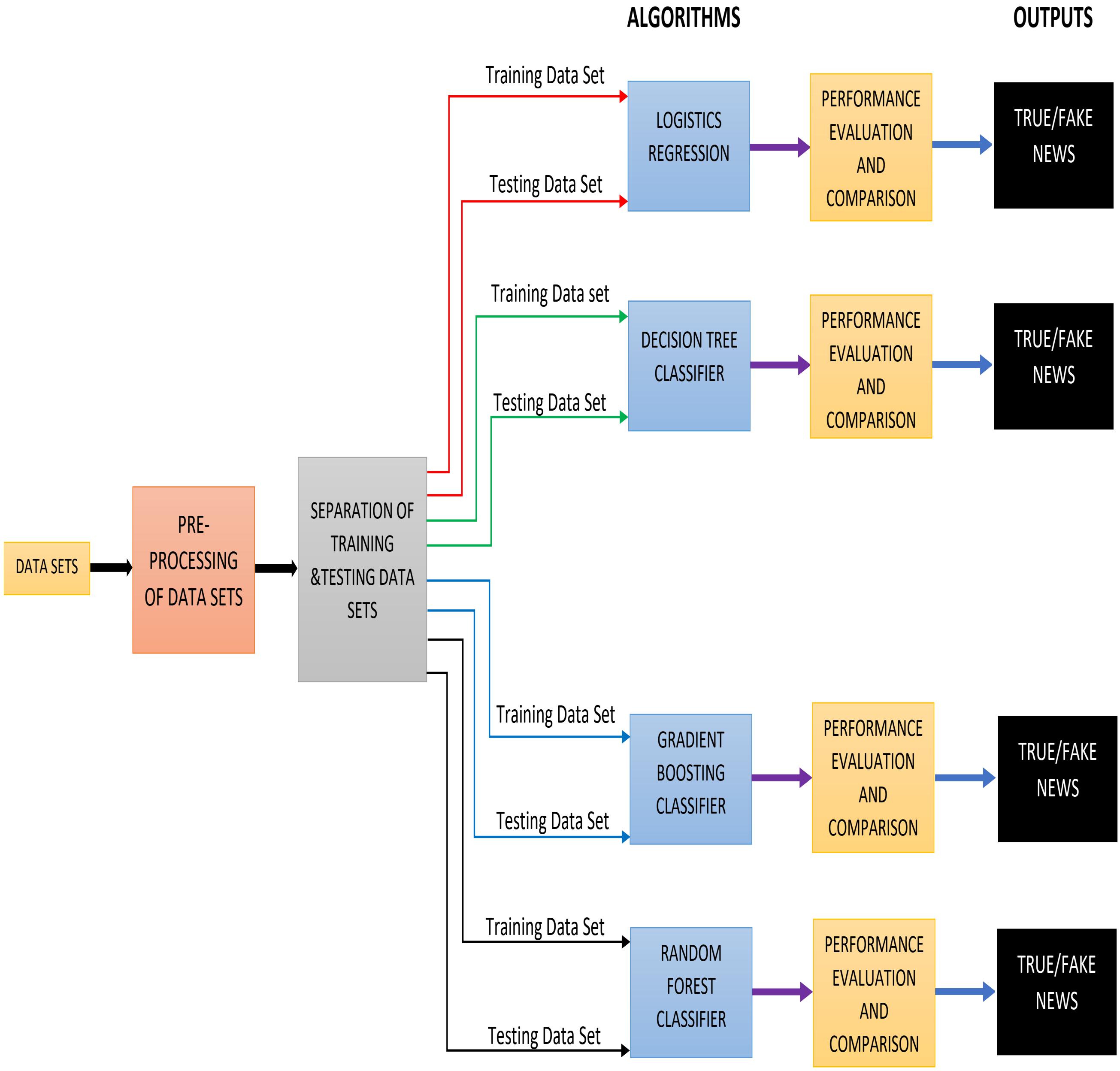

Figure:3.2: Block diagram of fake news detection

Datasets Have been loaded to build the model after reading the datasets. pre-processing of datasets takes place in which vectorisation technique is also involved. After pre-processing of datasets, we separate them into testing and training datasets. The training data is fit to all the algorithms to train and we pass the testing data to all the algorithms to predict the output. Performance of each algorithms is evaluated and compared with each other.

3.2.1 Datasets

The dataset contains two types of articles fake and real News. This dataset was collected from real world sources; the truthful articles were obtained by crawling articles from Reuters.com (News website). As for the fake news articles, they were collected from different sources. The fake news articles were collected from unreliable websites that were flagged by PolitiFact (a fact-checking organization in the USA) and Wikipedia. The dataset contains different types of articles on different topics; however, the majority of articles focus on political and World news topics.

The dataset consists of two CSV files. The first file named “True.csv” contains more than 12,600 articles from reuter.com. The second file named “Fake.csv” contains more than 12,600 articles from different fake news outlet resources. Each article contains the following information: article title, text, type and the date the article was published on. To match the fake news data collected for kaggle.com, we focused mostly on collecting articles from 2016 to 2017. The data collected were cleaned and processed, however, the punctuations and mistakes that existed in the fake news were kept in the text.

The following table gives a breakdown of the categories and number of articles per category.

Table 3.2.1 Datasets

News	Size (Number of articles)	Subjects	
Real-News	21417	**Type**	**Articles size**
		World-News	10145
		Politics-News	11272
Fake-News	23481	**Type**	**Articles size**
		Government-News	1570
		Middle-east	778
		US News	783
		left-news	4459
		politics	6841
		News	9050

To cite this dataset use

1. Ahmed H, Traore I, Saad S. "Detecting opinion spams and fake news using text classification", Journal of Security and Privacy, Volume 1, Issue 1, Wiley, January/February 2018.

2. Ahmed H, Traore I, Saad S. (2017) "Detection of Online Fake News Using N-Gram Analysis and Machine Learning Techniques. In: Traore I.,Woungang I., Awad A. (eds) Intelligent, Secure, and Dependable Systems in Distributed

and Cloud Environments. ISDDC 2017. Lecture Notes in Computer Science, vol 10618. Springer, Cham (pp. 127- 138).

3.2.2 Pre-Processing of Datasets

Social media data is highly unstructured – majority of them are informal communication with typos, slangs and bad-grammar etc. Quest for increased performance and reliability has made it imperative to develop techniques for utilization of resources to make informed decisions . To achieve better insights, it is necessary to clean the data before it can be used for predictive modelling. For this purpose, basic pre-processing was done on the News training data. This step was comprised of-Data Cleaning:

While reading data, we get data in the structured or unstructured format. A structured format has a well-defined pattern whereas unstructured data has no proper structure. In between the 2 structures, we have a semi-structured format which is a comparably better structured than unstructured format.

3.2.3 Separation of Training and Testing Data

- We Separate Our Datasets as Training and Testing in The Ratio Of 75:25
- Training Data Is the Data Which We Pass to The Algorithm to Train Them.
- Testing Data Is the Data Which We Test Them It Is the Unseen Data Which the Model Will Predict the Output.

Training Data-75%
Testing Data-25%

Vectorizing Data

Vectorizing is the process of encoding text as integers i.e. numeric form to create feature vectors so that machine learning algorithms can understand our data.

1. Vectorizing Data: Bag-Of-Words Bag of Words (Bow) or Count vectorizer describes the presence of words within the text data. It gives a result of 1 if present in the sentence and 0 if not present. It, therefore, creates a bag of words with a document-matrix count in each text document.

2. Vectorizing Data: N-Grams are simply all combinations of adjacent words or letters of length n that we can find in our source text. N-grams with n=1 are called unigrams. Similarly, bigrams (n=2), trigrams (n=3) and so on can also be used. Unigrams usually don't contain much information as compared to bigrams and trigrams. The basic principle behind n-grams is that they capture the letter or word is likely to follow the given word. The longer the n-gram (higher n), the more context you have to work with [20].

3. Vectorizing Data: TF-IDF It computes "relative frequency" that a word appears in a document compared to its frequency across all documents TF-IDF weight represents the relative importance of a term in the document and entire corpus [17].

TF stands for Term Frequency: It calculates how frequently a term appears in a document. Since, every document size varies, a term may appear more in a long sized document that a short one. Thus, the length of the document often divides Term frequency. Note: Used for search engine scoring, text summarization, document clustering.

$$TF(t,d) = \frac{\text{Number of times t occurs in document 'd'}}{\text{Total word count of documents 'd'}} \quad \text{.................(3.1)}$$

IDF stands for Inverse Document Frequency: A word is not of much use if it is present in all the documents. Certain terms like "a", "an", "the", "on", "of" etc. appear many times in a document but are of little importance. IDF weighs down the importance of these terms and increase the importance of rare ones. The more the value of IDF, the more unique is the word.

$$\mathrm{IDF(t,d)} = \frac{\text{Total number of documents}}{\text{Number of documents with term t in it}} \quad \text{..(3.2)}$$

TF-IDF is applied on the body text, so the relative count of each word in the sentences is stored in the document matrix.

$$TFIDF(t, d) = TF(t, d) * IDF(t) \quad \text{.........................(3.3)}$$

3.2.4 Algorithms

1. Logistic Regression

It is a classification not a regression algorithm. It is used to estimate discrete values (Binary values like 0/1, yes/no, true/false) based on given set of independent variable(s). In simple words, it predicts the probability of occurrence of an event by fitting data to a logit function. Hence, it is also known as logit regression. Since, it predicts the probability, its output values lies between 0 and 1 (as expected). Mathematically, the log odds of the outcome are modelled as a linear combination of the predictor variables .

Odds = p/(1-p) = probability of event occurrence / probability of not event occurrence

ln(odds) = ln(p/(1-p))

logit(p)=ln(p/(1-p))= b0+b1X1+b2X2+b3X3....+bkXk

2. Decision Tree Classifier

Decision Tree algorithm belongs to the family of supervised learning algorithms. Unlike other supervised learning algorithms, the decision tree algorithm can be used for solving **regression and classification problems** too. The goal of using a Decision Tree is to create a training model that can use to predict the class or value of the target variable by **learning simple decision rules** inferred from prior data(training data).

In Decision Trees, for predicting a class label for a record we start from the **root** of the tree. We compare the values of the root attribute with the record's attribute. On the basis of comparison, we follow the branch corresponding to that value and jump to the next node.

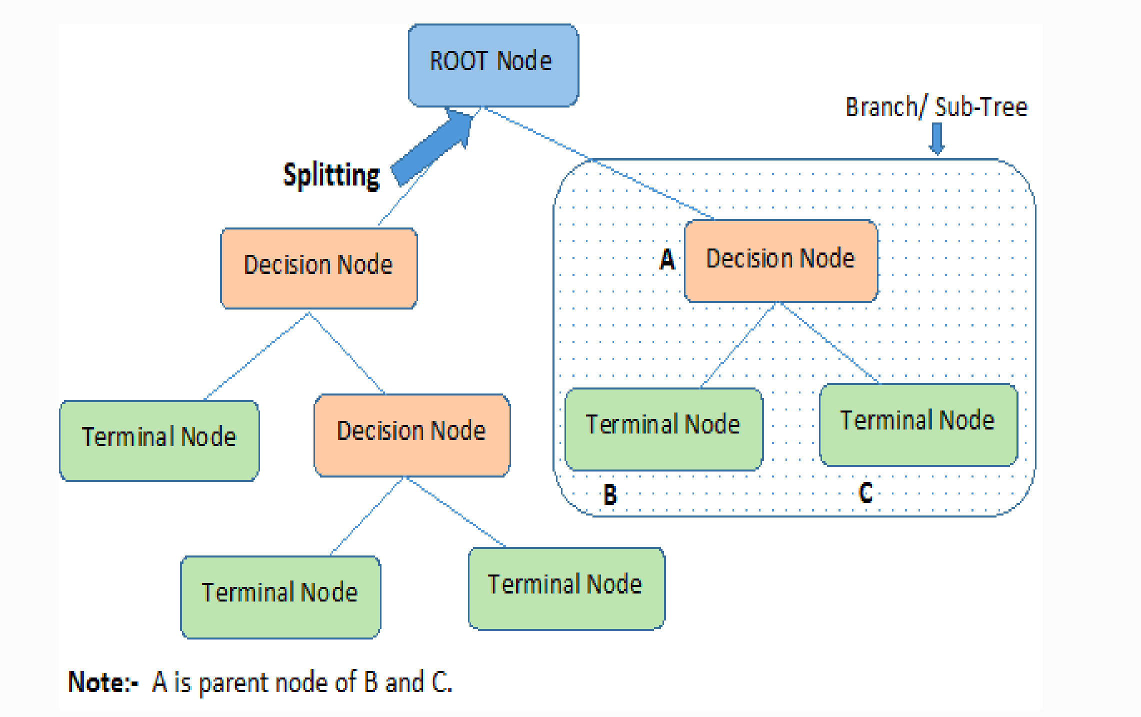

Figure: Model of Decision Tree Classifier

Types of Decision Trees

1. **Root Node:** It represents the entire population or sample and this further gets divided into two or more homogeneous sets.
2. **Splitting:** It is a process of dividing a node into two or more sub-nodes.
3. **Decision Node:** When a sub-node splits into further sub-nodes, then it is called the decision node.
4. **Leaf / Terminal Node:** Nodes do not split is called Leaf or Terminal node.
5. **Pruning:** When we remove sub-nodes of a decision node, this process is called pruning. You can say the opposite process of splitting.
6. **Branch / Sub-Tree:** A subsection of the entire tree is called branch or sub-tree.
7. **Parent and Child Node:** A node, which is divided into sub-nodes is called a parent node of sub-nodes whereas sub-nodes are the child of a parent node.

3. Gradient boosting classifier

Gradient boosting is a machine learning technique for regression, classification and other tasks, which produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees. When a decision tree is the weak learner, the resulting algorithm is called gradient boosted trees, which usually outperforms random forest. It builds the model in a stage-wise fashion like sother boosting methods do, and it generalizes them by allowing optimization of an arbitrary differentiable loss function.

The Python machine learning library, Scikit-Learn, supports different implementations of gradient boosting classifiers, including XGBoost.

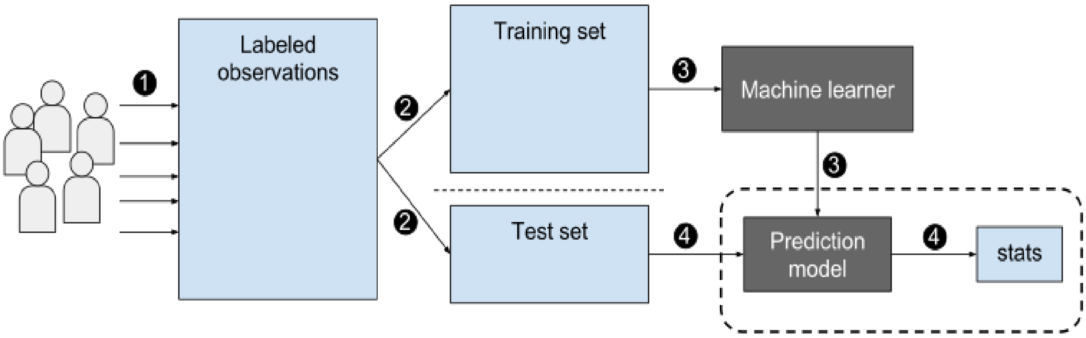

Figure: Model of Gradient boosting classifier

In order to implement a gradient boosting classifier, we'll need to carry out a number of different steps. We'll need to:

- Fit the model
- Tune the model's parameters and Hyperparameters
- Make predictions
- Interpret the results

4. Random forest classifier

Random Forest is a trademark term for an ensemble of decision trees. In Random Forest, we've collection of decision trees (so known as "Forest"). To classify a new object based on attributes, each tree gives a classification and we say the tree "votes" for that class. The forest chooses the classification having the most votes (over all the trees in the forest). The random forest is a classification algorithm consisting of many decisions trees. It uses bagging and feature randomness when building each individual tree to try to create an uncorrelated forest of trees whose prediction by committee is more accurate than that of any individual tree. Random forest, like its name implies, consists of a large number of individual decision trees that operate as an ensemble.

Each individual tree in the random forest spits out a class prediction and the class with the most votes becomes our model's prediction. The reason that the random forest model works so well is: A large number of relatively uncorrelated models (trees) operating as a committee will outperform any of the individual constituent models. So how does random forest ensure that the behaviour of each individual tree is not too correlated with the behaviour of any of the other trees in the model.

It uses the following two methods:

1. **Bagging (Bootstrap Aggregation)** — Decisions trees are very sensitive to the data they are trained on small changes to the training set can result in significantly different tree structures. Random forest takes advantage of this by allowing each individual tree to randomly sample from the dataset with replacement, resulting in different trees. This process is known as bagging or bootstrapping.

2. **Feature Randomness** — In a normal decision tree, when it is time to split a node, we consider every possible feature and pick the one that produces the most separation between the observations in the left node vs. those in the right node. In contrast, each tree in a random forest can pick only from a random subset of features. This forces even more variation amongst the trees in the model and ultimately results in lower correlation across trees and more diversification.

3.2.5 Evaluation Metrics

Evaluate the performance of algorithms for fake news detection problem; various evaluation metrics have been used. In this subsection, we review the most widely used metrics for fake news detection. Most existing approaches consider the fake news problem as a classification problem that predicts whether a news article is fake or not:

True Positive (TP): when predicted fake news pieces are actually classified as fake news;

True Negative (TN): when predicted true news pieces are actually classified as true news;

False Negative (FN): when predicted true news pieces are actually classified as fake news;

False Positive (FP): when predicted fake news pieces are actually classified as true news.

3.2.6 Confusion Matrix

A confusion matrix is a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known. It allows the visualization of the performance of an algorithm. A confusion matrix is a summary of prediction results on a classification problem. The number of correct and incorrect predictions are summarized with count values and broken down by each class. This is the key to the confusion matrix. The confusion matrix shows the ways in which your classification model is confused when it makes predictions. It gives us insight not only into the errors being made by a classifier but more importantly the types of errors that are being made.

Table 3.2.6: confusion matrix

Total	Class 1 (Predicted)	Class 2 (Predicted)
Class 1 (Actual)	TP	FN
Class 2 (Actual)	FP	TN

By formulating this as a classification problem, we can define following metrics

1. Precision $=\frac{|T\,P|}{|T\,P|+|F\,P|}$ ………………………………….(3.4)
2. Recall $= \frac{|T\,P|}{|TP|+|FN|}$ ………………………………….(3.5)
3. F1 Score $= 2 \text{ x} \frac{\text{Precisionn} * \text{Recall}}{\text{Precision} + \text{Recall}}$ ………………………(3.6)
4. Accuracy $= \frac{|T\,P|+|T\,N|}{|T\,P|+|TN|+|F\,P|+|F\,N|}$ ……………………..(3.7)

These metrics are commonly used in the machine learning community and enable us to evaluate the performance of a classifier from different perspectives. Specifically, accuracy measures the similarity between predicted fake news and real fake news

Recall actually calculates how many of the Actual Positives our model capture through labelling it as Positive (True Positive). Applying the same understanding, we know that Recall shall be the model metric we use to select our best model when there is a high cost associated with False Negative.

Chapter 4

IMPLEMENTATION DETAILS

4.1 Coding Language Used For Implementation

PYTHON- Python is an easy to learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. The Python interpreter and the extensive standard library are freely available in source or binary form for all major platforms from the Python Web site, https://www.python.org and may be freely distributed. The same site also contains distributions of and pointers to many free third-party Python modules, programs and tools, and additional documentation. The Python interpreter is easily extended with new functions and data types implemented in C or C++ (or other languages callable from C). Python is also suitable as an extension language for customizable applications.

4.2 History Of Python

1. Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.
2. Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, Small Talk, and Unix shell and other scripting languages.
3. Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).
4. Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

4.3 Pythons Features

1. Easy-to-learn − Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.

2. Easy-to-read − Python code is more clearly defined and visible to the eyes.

3. Easy-to-maintain − Python's source code is fairly easy-to-maintain.

4. A broad standard library − Python's bulk of the library is very portable and crossplatform compatible on UNIX, Windows, and Macintosh.

5. Interactive Mode − Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.

6. Portable − Python can run on a wide variety of hardware platforms and has the same interface on all platforms.

7. Extendable − You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.

8. Databases − Python provides interfaces to all major commercial databases.

9. GUI Programming − Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.

10. Scalable − Python provides a better structure and support for large programs than shell scripting.

11. It supports functional and structured programming methods as well as OOP.

12. It can be used as a scripting language or can be compiled to byte-code for building large applications.

13. It provides very high-level dynamic data types and supports dynamic type checking.

14. It supports automatic garbage collection.

15. It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

4.4 Software Used For Implementation

Anaconda- Anaconda is a free and open-source distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment. Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda ® distribution that allows you to launch applications and easily manage anaconda packages, environments, and channels without using commandline commands. Navigator can search for packages on Anaconda Cloud or in a local Anaconda Repository. It is available for Windows, macOS, and Linux.

4.5 Applications Of Anaconda

1. Jupyter Lab
2. Jupyter Notebook
3. Spyder
4. PyCharm
5. VSCode
6. Glueviz
7. Orange 3 App
8. RStudio
9. Anaconda Prompt (Windows only)
10. Anaconda PowerShell (Windows only)

4.6 The Jupyter Notebook

1. The notebook extends the console-based approach to interactive computing in a qualitatively new direction, providing a web-based application suitable for capturing the whole computation process: developing, documenting, and executing code, as well as communicating the results. The Jupyter notebook combines two components:

2. A web application: a browser-based tool for interactive authoring of documents which combine explanatory text, mathematics, computations and their rich media output.

3. Notebook documents: a representation of all content visible in the web application, including inputs and outputs of the computations, explanatory text, mathematics, images, and rich media representations of objects. These documents are internally JSON files and are saved with the ipynb extension. Since JSON is a plain text format, they can be version-controlled and shared with colleagues.

4. Notebooks may be exported to a range of static formats, including HTML (for example, for blog posts), restructured text, LaTeX, PDF, and slide shows, via the nbconvert command. Any .ipynb notebook document available from a public URL can be shared via the Jupyter Notebook Viewer. This service loads the notebook document from the URL and renders it as a static web page. The results may thus be shared with a colleague, or as a public blog post, without other users needing to install the Jupyter notebook themselves.

4.7 Flowchart

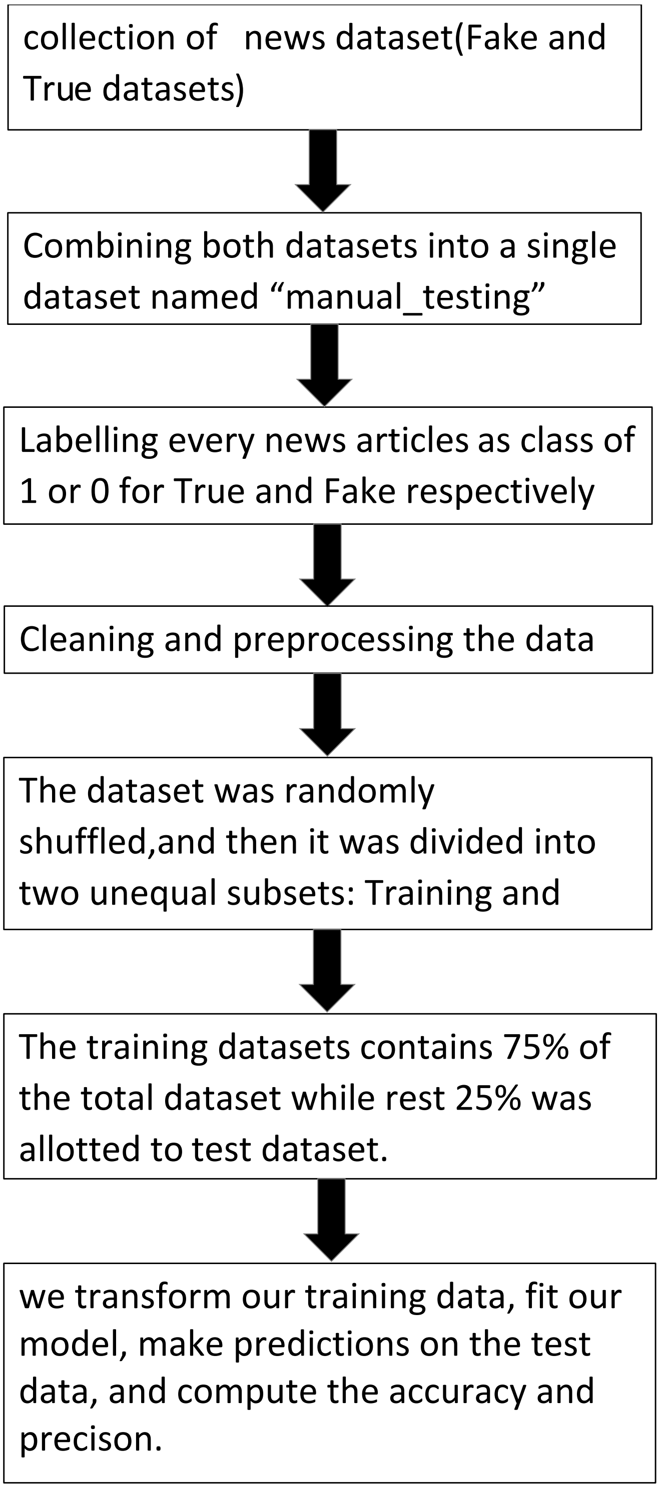

Figure: Flowchart of fake news detection

Chapter 5

RESULTS

Out[9]:

	title	text	subject	date	class
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017	0
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017	0
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017	0
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017	0
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017	0
5	Racist Alabama Cops Brutalize Black Boy While...	The number of cases of cops brutalizing and ki...	News	December 25, 2017	0
6	Fresh Off The Golf Course, Trump Lashes Out A...	Donald Trump spent a good portion of his day a...	News	December 23, 2017	0
7	Trump Said Some INSANELY Racist Stuff Inside ...	In the wake of yet another court decision that...	News	December 23, 2017	0
8	Former CIA Director Slams Trump Over UN Bully...	Many people have raised the alarm regarding th...	News	December 22, 2017	0
9	WATCH: Brand-New Pro-Trump Ad Features So Muc...	Just when you might have thought we d get a br...	News	December 21, 2017	0

Fig 5.1 dataset before preprocessing

The above figure 5.1 shows the output of datasets before pre-processing where null values are checked, punctuations marks, URLs, upper case letters are present. ‘0’ indicates fake news. ‘date’ and ‘subject’ column is not required therefore we are going to drop this column in the next process.

Out[16]:

	text	class
10919	in its year history only one f b i director...	0
12335	the president who shoved his radical agenda do...	0
8325	everett washington washington reuters rep...	1
19742	wikileaks released an email from center for am...	0
926	donald trump s obsession with twitter and inab...	0
20280	moscow reuters russian defence minister se...	1
5768	as polls show clinton beating donald trump in ...	0
8685	cleveland reuters republican presidential ...	1
17884	london reuters brexit is not a game the...	1
17368	kuala lumpur reuters the battlefield death...	1

Fig 5.2 dataset after preprocessing

The above Figure.5.2. shows the output of datasets after pre-processing where punctuations marks, URLs, upper case letters are removed. date’ and ‘subject’ column has been removed.’1’ indicates true news detection.

Logistic Regression

```
In [21]: from sklearn.linear_model import LogisticRegression

In [22]: LR=LogisticRegression()
         LR.fit(xv_train,y_train)

Out[22]: LogisticRegression()

In [23]: LR.score(xv_test,y_test)

Out[23]: 0.9844919786096257

In [24]: pred_LR=LR.predict(xv_test)

In [25]: print(classification_report(y_test,pred_LR))

              precision    recall  f1-score   support

           0       0.99      0.98      0.99      5864
           1       0.98      0.99      0.98      5356

    accuracy                           0.98     11220
   macro avg       0.98      0.98      0.98     11220
weighted avg       0.98      0.98      0.98     11220
```

Fig 5.3 Logistics regression output

The above figure 5.3 shows the output of logistics regression(LR) where we fit the training data into the LR model. We pass the testing data and predict the output accuracy obtained was 98.4%.

```
In [26]: from sklearn.tree import DecisionTreeClassifier

In [27]: DT=DecisionTreeClassifier()
         DT.fit(xv_train,y_train)

Out[27]: DecisionTreeClassifier()

In [28]: DT.score(xv_test,y_test)

Out[28]: 0.9948306595365419

In [29]: pred_DT=DT.predict(xv_test)

In [30]: print(classification_report(y_test,pred_DT))

              precision    recall  f1-score   support

           0       0.99      1.00      1.00      5864
           1       0.99      0.99      0.99      5356

    accuracy                           0.99     11220
   macro avg       0.99      0.99      0.99     11220
weighted avg       0.99      0.99      0.99     11220
```

Fig 5.4 Decision tree classifier output

The above figure 5.4 shows the output of Decision tree classifier(DT) where we fit the training data into the DT model. We pass the testing data and predict the output accuracy obtained was 99.4%.

###Gradient Boosting Classifier

```
In [31]: from sklearn.ensemble import GradientBoostingClassifier

In [32]: GBC=GradientBoostingClassifier(random_state=0)
         GBC.fit(xv_train, y_train)

Out[32]: GradientBoostingClassifier(random_state=0)

In [33]: GBC.score(xv_test,y_test)

Out[33]: 0.9950089126559715

In [34]: pred_GBC=GBC.predict(xv_test)

In [35]: print(classification_report(y_test,pred_GBC))
                       precision    recall  f1-score   support

                    0       1.00      0.99      1.00      5864
                    1       0.99      1.00      0.99      5356

             accuracy                           1.00     11220
            macro avg       0.99      1.00      0.99     11220
         weighted avg       1.00      1.00      1.00     11220
```

Fig 5.5 Gradient boosting classifier output

The above figure 5.4 shows the output of Gradient boosting classifier(GB) where we fit the training data into the GB model. We pass the testing data and predict the output accuracy obtained was 99.5%.

###Random Forest Classifier

```
In [36]: from sklearn.ensemble import RandomForestClassifier

In [37]: RFC=RandomForestClassifier(random_state=0)
         RFC.fit(xv_train,y_train)

Out[37]: RandomForestClassifier(random_state=0)

In [38]: RFC.score(xv_test,y_test)

Out[38]: 0.9899286987522281

In [39]: pred_RFC=RFC.predict(xv_test)

In [40]: print(classification_report(y_test,pred_RFC))
                      precision    recall  f1-score   support

                   0       0.99      0.99      0.99      5864
                   1       0.99      0.99      0.99      5356

            accuracy                           0.99     11220
           macro avg       0.99      0.99      0.99     11220
        weighted avg       0.99      0.99      0.99     11220
```

Fig 5.6 Random forest classifier output

The above figure 5.4 shows the output of Random forest classifier(RF) where we fit the training data into the RF model. We pass the testing data and predict the output accuracy obtained was 98.9%.

Comparison

Comparing all the four algorithms we got the highest accuracy of Gradient Boosting Algorithm with the accuracy of 99.5%

Chapter 6

CONCLUSION

In this mini project work, we have addressed the problem of classifying fake news articles using machine learning models and ensemble techniques was implemented. The learning models were trained and tested to obtain optimal accuracy. We used multiple performance metrics to compare the results for each algorithm. We have discussed the various ways, methods and models of fake news detection. We have addressed the benefits and shortcomings of each model. We have given the comparison of accuracy levels of all the models. We plan to overcome the shortcomings of these models through our project. We will make use of two datasets, a full training dataset and a testing training data set, associated with news articles. These datasets are obtained from Kaggle. We will use machine Language Processing to detect fake news directly. Finally, we concluded that we got the highest accuracy of 99.5% in Gradient Boosting Algorithm.

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